

wx \ yz	00	01	11	10
00		1	1	
01	1	1		
11		1		
10		1	1	1

wx \ yz	00	01	11	10
00		1	1	
01	1	1		
11	1	1		
10		1	1	1

Simplified: $xy' + x'z + wx'y$

3-5: Given the following truth table:

x	y	z	F ₁	F ₂
0	0	0	0	0
0	0	1	1	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

(a) Express F₁ and F₂ in product of maxterms

$F_1 = (x+y+z) \cdot (x+\bar{y}+\bar{z}) \cdot (\bar{x}+\bar{y}+z)$

$F_2 = (x+y+z) \cdot (x+y+\bar{z}) \cdot (x+\bar{y}+z) \cdot (\bar{x}+y+z)$

(b) Obtain the simplified functions in sum of products.

Simplified F₁: Original F₁
 $= xy'z + x'yz' + xy'z'$

F₁

wx \ yz	00	01	11	10
00		1		1
01	1		1	
11		1		
10		1	1	1

F₂

wx \ yz	00	01	11	10
00			1	
01		1	1	
11		1	1	1
10		1	1	1

Simplified F₂: $yz + xy + xz$